

Abstract

Purpose of the Study

The purpose of this study is to design a secure, intelligent, and real-time framework that can automatically detect fraud and make reliable decisions with minimal human involvement. The study focuses on combining real-time data engineering with self-correcting agentic AI and quantum-safe security mechanisms to ensure fast, trustworthy, and future-ready autonomous decision-making in digital systems.

Background and Context

Modern digital platforms such as online banking, payment applications, and e-commerce systems generate huge volumes of transaction data every second. Traditional batch-based systems process data with delays and depend heavily on manual monitoring, which often results in late fraud detection and higher financial losses. At the same time, the rise of autonomous AI systems raises concerns about trust, accountability, and long-term security, especially with the future threat of quantum computing. This creates a strong need for systems that are real-time, autonomous, transparent, and quantum-safe.

Methodology

The proposed framework is built on real-time data engineering pipelines that process live transaction data instantly. Autonomous AI agents continuously monitor data streams, detect anomalies, correct their own decision errors, and adapt to new fraud patterns without manual intervention. A quantum-safe logging and auditing layer securely records every AI decision, ensuring traceability and resistance to future quantum attacks. The architecture is fully software-based, distributed, and scalable.

Key Findings

The framework significantly reduces detection latency, improves fraud identification accuracy, and enhances trust in autonomous decisions. Self-correcting agents reduce manual workload, while quantum-safe logs ensure transparency and long-term security.

Future Research

Future work may include advanced learning agents, real-world deployment testing, and deeper integration of post-quantum cryptographic techniques for large-scale autonomous systems.